

DRUG OVERDOSE; STUDY CONDUCTED AT NATIONAL POISON CONTROL CENTER

AFTAB TURABI, SAUD HASAN, JAMAL ARA AND MANSOOR AHMED

*Department of Pharmacology & Therapeutics
Islamic International Medical College, 274, Peshawar Road, Rawalpindi*

ABSTRACT:

A prospective study of 588 cases of drug overdose was carried out in National Poison Control Centre (NPCC), Jinnah Postgraduate Medical Centre Karachi (JPMC) from 1999-2002. It was done to determine the incidence of renal, hepatic, respiratory, neural and cardiac injuries or toxicity. In this study all age groups were included above 8 years. Various investigations were carried out that include blood CP. sugar, electrolytes, urea, creatinine, blood gases, electrocardiogram, ultrasound and Urine examination. The drug Overdose cases admitted from 1999-2002 were exposed to various drugs, such as benzodiazepines, barbiturates, non-steroidal anti-inflammatory drugs (NSAID), beta-blockers, carbamezapine, and Vitamin-A. The mortality rate was 2.89% with a male /female ratio of 76.47%: 23.53%. However the recovery was 94.22%. These drug overdose cases were found to be the consequences of suicidal, accidental or homicidal attempts.

This study may help in determining the goals for reducing the morbidity and mortality of drug over dose as it shows the rate of incidence, gender ratio, clinical presentations, diagnostic evaluations, complications, management, outcome and medicolegal aspects.

INTRODUCTION

National Poison Control Centre (NPCC) Karachi, is working since 1984. This is the 1st detailed report on the overdose aspect of drugs. Internationally drug overdose constitutes a large group among various categories of poisoning. The Patients admitted in National Poison Control Centre (NPCC) – Jinnah Postgraduate Medical Centre, Karachi are usually coming from the whole Sindh province and partly from Baluchistan. Usually the patient reaches the center within 24 to 72 hours of the exposure of drug overdose depending upon the distance, awareness and the education of the people. In this article we are reporting the overdose toxicity of acetaminophen, salicylic acid, benzodiazepines and Vitamin 'A'.

The literature search on these groups internationally and nationally are quite alarming and need special attention because of the increased rate of morbidity and mortality

in this group of poisoning. Since the establishment of NPCC (1984 till present) the rate of poisoning has been increasing every year. This article is written with the objective of data analysis, types of clinical presentation and management given its pit falls and to determine the ultimate outcome of drug overdose. This study can provide guideline for the treatment of drug overdose and create an awareness among public, medical and allied personnel that in turn will help in decreasing the rate of its morbidity and mortality.

METHOD

The classical established method was used in care taking and treatment/management of drug overdose. 588 patients were admitted to the NPCC during past four years (1999 to 2002) with a history of drug overdose. The drugs included in this study were acetaminophen, salicylic acid (aspirin), benzodiazepines, barbiturates, beta-blockers, carbamezapine, NSAIDS and vitamin 'A'.

Table-1
Drugs taken by Patients from January 1999 to December 2002

Drugs	No. of Patients per year Drugs				
	1999	2000	2001	2002	Total
Benzodiazepine	69	81	54	73	277
Barbiturates	37	25	23	29	114
β -Blockers	11	07	04	05	27
Carbamezapine	02	06	04	01	13
NSAIDS	28	56	21	49	154
Vitamin A	0	0	01	02	03

Table-2
Duration between the Drug Intake and Admission in NPCC

Drugs	<6 hours	>12 hours	>24 hours	>36-48 hr	> 48-72 hr
Acetaminophen	-	-	01	02	02
Benzodiazepine	21	32	36	29	06
Barbiturates	17	19	14	07	-
β -Blockers	02	04	05	-	-
Carbamezapine	03	06	04	-	-
NSAIDS	19	13	12	02	-
Vitamin A	-	-	01	-	-

Table-3
Route of Administration in Drug Overdose from January 1999 to December 2002

Route of Administration	Male		Female		Total	
	Number	Percentage	Number	Percentage	Number	Percentage
Ingestion	281	47.79%	307	52.21%	588	100%
Inhalation	0	0	0	0	0	0
Parenteral	0	0	0	0	0	0
Dermal	0	0	0	0	0	0

These cases were admitted with different clinical presentations ranging from palpitations, drowsiness, abdominal pain, nausea vomiting, circulatory collapse, respiratory distress, gastrointestinal bleed and coma of all grades. The diagnosis was made on the basis of history, clinical evaluation and laboratory investigations. All patients received immediate ICU care that included vitals monitoring, gastric lavage and maintenance of IV line. Specific management was given

varying from case to case. Complications if developed were treated accordingly.

The biochemical investigations performed were blood CP, urine examination, blood sugar, electrolytes, blood urea, serum creatinine, liver function test; prothrombin time, APPT and blood gases.

Patient's gender, age group, Modus operandi, treatment; complications, medico-

Table-4
Outcome in Drug Overdose from January 1999 to December 2002
(n=588)

Outcome	Male		Female		Total	
	Number	Percentage	Number	Percentage	Number	Percentage
Recovery	262	47.29%	292	52.71%	554	94.22%
Death	13	76.47%	4	23.53%	17	2.89%
LAMA	6	35.29%	11	64.71%	17	2.89%

Table-5
Medico Legal Data of Drug Overdose from January 1999 to December 2002
(n=588)

	Male		Female		Total	
	Number	Percentage	Number	Percentage	Number	Percentage
Suicidal	163	43.12%	215	56.88%	378	64.29%
Accidental	118	56.19%	92	43.81%	210	35.71%
Homicidal	0	0	0	0	0	0

legal status and outcome were studied in every case.

RESULTS

The four-year work showed that out of 4894 cases of poisoning 588 cases were of drug overdose that constituted (17.63%. The male female difference was 281: 307 (47.79%: 52.21%). Regarding medico legal status, the suicidal group had 378(64.29%) patients out of which 163 were male (43.12%) and 215 were females 56.88%). The second on the list was accidental group that had 210 patients (35.71%) out of which 118 (56A9%) were males and 92(43.81%) were females. There was not a single case of homicidal poisoning reported (Table 5).

In this study of drug overdose the only route of exposure was found to be ingestion i.e. 100% (Table 3).

The outcome of drug overdose resulted in 17 deaths (2.89 %) had male/female ratio of 76.47% to 23.53%. The patients who left the

ward against medical advice (LAMA) were also 17 i.e. 2.89% and had male/female difference of 6:11 (35.29%: 64.71%). There was a complete recovery of 554 patients i.e. 9422% with male/female population ratio of 262: 292 i.e. 47.29%: 52.71% (Table-4).

Various drugs identified in this group during the period of 4 years included benzodiazepines (277), non-steroidal anti inflammatory drugs (154), barbiturates (114), beta-blockers (27), carbamazepine (13), and vitamin A (3) (Table 1 and Fig. 1). The common NSAID'S were salicylic acid and acetaminophen, whereas others were ibuprofen. Diclofenac sodium, mefenamic acid and piroxicam (Fig. 2). Benzodiazepines encountered during the study were diazepam (valium), lorazepam (ativan), xanax (alprazolam), and bromazepam (lexatonil), etc. available in different strength of 0.2 mg to 6 mg per tablet (Fig. 3).

DISCUSSION

Drugs were found to be the third most

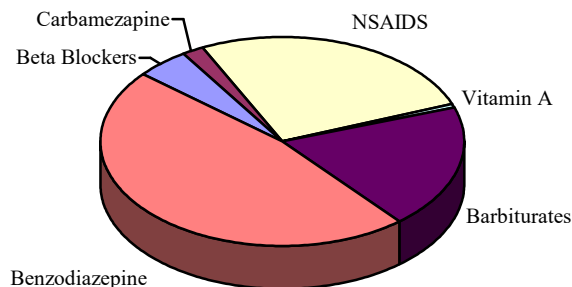


Fig. 1: Different drugs taken from 1999 to 2002.

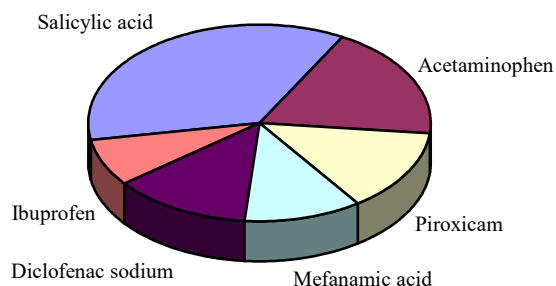


Fig. 2: NSAIDS taken from January 1999 to December 2002 (n=154).

common cause of poisoning leading to death in 1995 the USA. It is further stated that drug overdose is the main cause of death in 35-44 years age group. However 59% of poisoning cases occur in the age group below 14 years. Literature search showed the common drugs that cause poisoning are tranquilizers and antipsychotics, especially benzodiazepines, phenothiazines and tricyclic antidepressants (Fingerhut & Cox, 1998 and Jaraczewska *et al.*, 1997).

In our study the drug overdose constituted 17.63% of the total poisoning cases and comes next to the pesticide poisoning. The common drugs that we came across were benzodia-

zepines salicylic acids acetaminophen. These drugs are freely available to people, as usually they do not require any medical prescription, although the legislation or drug act is there. According to the act no drug should be sold or purchased without medical prescription but unfortunately no body follows the law. By educating the chemist and limiting the number of tablets dispensed at a time may help in reducing the incidences of drug overdose in Pakistan. Above all government and health ministry should take serious steps to implement the law.

Benzodiazepines were found to be the most commonly used medicine among urban

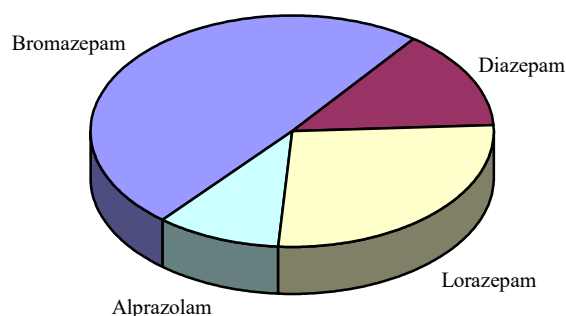


Fig. 3: Benzodiazepines taken from January 1999 to December 2002.

population Its free availability in almost every house made easy accessibility by children and teenagers. They are now the most commonly prescribed drugs in many countries. No other group of drugs has proved as versatile as the benzodiazepines, probably because of their wide margin of safety that also encourages over prescription and misuse. The benzodiazepines are often prescribed as a remedy for pressures and problems of life in people who are disappointed, unhappy, or frustrated. Mortality with benzodiazepines is very rare (Khan & Reza, 1998), although our study showed 4 deaths because of evident benzodiazepine over dose. These 4 cases were brought to NPCC in a state of deep coma with central respiratory depression The history revealed that they were found unconscious in their room or house probably for longer duration i.e. more than 24 hours. They may have taken some other drugs or substance concomitantly, which remained unrecognized. The high dose and longer duration of exposure leads to irreversible brain damage. Urine was found positive for benzodiazepines with high cutoff value of ≥ 450 in these 4 patients. Flumazenil, given up to 2 mg, was not able to reverse the central depressive effects in these 4 cases, although it proves its efficacy in other cases of benzodiazepine over dose. The majority responded to the single dose (0.3 μg)

of flumazenil, however in some cases multiple dose or continuous infusion was used. Placebo-controlled double-blind studies have confirmed the beneficial effects of flumazenil in cases of benzodiazepine poisoning (Aaseth *et al.*, 1988; Bismuth *et al.*, 1985 and Geller *et al.*, 1985).

It is stated that if respiratory depression is mediated through the benzodiazepine receptor, then flumazenil should be effective as a reversal agent, but this effect does not occur consistently (Lim, 1989; Morga *et al.*, 1989 and Shalansky *et al.*, 1993). Flumazenil awakened patient rapidly but failed to affect minute ventilation volumes and had little effect on oxygen saturation. Rebound respiratory depression and prolonged hypoxic episodes were documented. Benzodiazepine-induced apnea should be managed with fundamental procedures such as supplemental oxygen, airway stabilization, bag-valve mask ventilation, and endotracheal intubation.

Paracetamol (Acetaminophen) is an important non-steroidal anti-inflammatory drug frequently used as antipyretic and analgesic. The fatal dose is ≥ 150 mg/kg. Six deaths have been recorded during four years study due to acetaminophen poisoning.

It is stated that once the damage becomes extensive the prognosis is very poor (Canalese *et al.*, 1981 and Clark & Thompson, 1973). In our study all 6 (20.69%) patients who were died due to Paracetamol overdose had hametemesis, hepatic encephalopathy, renal shut down and respiratory acidosis Other 23 (79.31%) patients recovered completely as they were brought to NPCC within 24 hours of Paracetamol overdose whereas the above mentioned 6 patients reported 36-48 hours after the intake (Table 2). This shows that if the victim of Paracetamol reaches the hospital or Poison control center within 24 hours he may survive as centrilobular necrosis of liver could not take place in this time, which is an irreversible condition. Different studies revealed that Paracetamol metabolism involves its conversion by cytochrome P-450-dependent mixed function oxidase to a reactive metabolite, N-acetyl-p-benzoquinone imine (NAPQI), which may cause acute hepatic necrosis with toxic doses (Dahlin *et al.*, 1984; Holme *et al.*, 1984 and Tirmenstein & Nelson, 1989). NAPQI causes depletion of both the mitochondrial and cytosolic pools of reduced glutathione (GSH) (Trazepacz *et al.*, 1992). The maintenance of hepatic glutathione (GSH) concentrations by administration of N-acetylcysteine was first suggested as a treatment for Paracetamol poisoning by (1995; Prescott *et al.*, 1974). However, GSH precursors such as N-acetylcysteine have been found to be effective both in experimental animals and humans (Boobis *et al.*, 1989). This was found appropriate in our study too, resulted in 79.31% recovery rate whereas 20.69% patients received the specific antidote after the development of hepatic necrosis as they arrived at the center after 36 hours of intake of Paracetamol.

Salicycic acid (Aspirin) is available in the form of tablets ointment. The fatal complications observed during the study in salicylate poisoning were haemetemesis, intestinal perforation and renal failure that lead to 7 (12.73%) deaths. Significant alteration in serum electrolytes was observed in 39 patients (70.9%), which was corrected during their stay

at NPCC.

Salicylate poisoning is not merely compensatory for the metabolic acidosis or vice versa, but acutely poisoned adult patient characteristically presents with a mixed acid-base disturbance that is discernible chiefly by astute ABG and serum electrolyte analysis. Salicylate manifest acute toxicity at the serum level of > 40 mg/dl in most cases but it can vary up to 100 mg/dl in certain individuals. The fatal dose is 200-600 mg/kg (2000; Goldfrank, 1994).

In our study only three cases of hypervitaminosis A were reported over a period of 4 years, among which one was male and two were females and were in the age group of 15-20 years. They received vitamin-A capsules during door-to-door campaign for Polio prevention program conducted by the government. Social worker banded them plenty of capsules, while simultaneously telling the beneficial effects of capsules especially on vision As all 3 of them were wearing eyeglasses they took 20-25 capsules as a single dose. After 6 hours of ingestion they developed nausea and headache followed by vomiting. They seek medical advice from local clinics where they were treated symptomatically and sent home. Overnight the condition was worsened and ultimately became unconscious in the morning when they were brought to NPCC. Examination revealed dried and desquamated skin, bone tenderness mainly over tibia, conjunctival hemorrhage, and signs of raised intra-cranial pressure. All of them responded well to the treatment and discharged after 3-5 days. Out patient follow-ups could not reveal any permanent sequel.

These types of incidences need special attention and the workers who are hired for these campaigns should receive proper training prior to the start of campaign it is also suggested that public education should be mandatory through press and electronic media during such programs.

REFERENCES

- Aaseth, H.P., Bredesen, J.E., Grynne, B., Lyngdal, P.T., Storstein, L. and Wiik-Larsen, E. (1988). Benzodiazepine-receptor antagonist, a clinical double blind study. *J. Toxicol. Clin. Toxicol.* **26**: 283-292.
- Antidotes for poisoning by Paracetamol, IPCS/EC Evaluation of Antidotes Series, Volume 3, 1995.
- Bismuth, C., Baud, F.J., Fournier, P.E., Mellerio, F. (1985). Antagoniste Ro-15-1788 dans l'intoxication volontaire par benzodiazepines. Valeur diagnostique et therapeutique. Etude randomisee sur 40 cas. *J. Toxicol. Clin. Exp.* **5**: 403404 (Abstract).
- Boobis, A.R., Fawthrop, D.J. and Davis, D.S. (1989). Mechanism of cell deaths. *Trends Pharmacol Sci.*, **10**: 275-280.
- British National Formulary. BNF 40.2000.
- Canalese, J., Gimson, A.E.S., Davies, M. and Williams R. (1981). Factors contributing to mortality in paracetamol-induced hepatic failure. *BMJ.* **282**: 199-201.
- Clark, R., Thompson, R.P.H., Borirakchan-yavat, V., Widdop, B., Davidson, A.R., Goulding, R. and Williams, R. (1973). Hepatic damage and death from overdose of paracetamol. *Lancet.* **1**: 66-70.
- Dahlin, D.C., Miwa, G.T., Lu, A.Y.H. and Nelson, S.D. (1984). N-acetyl-p-benzoquinone imine: a cytochrome P-450 mediated oxidation product of acetaminophen. *Proc. Natl. Acad. Sci. (USA)*. **81**: 1327-1331.
- Fingerhut, L.A. and Cox, C.S. (1998). Poisoning mortality, 1985-1995. *Public Health Rep.* **113**(3): 218-233.
- Geller, E. and Niv, M. (1985). Ro 15-1788, a benzodiazepine antagonist in the treatment of 34 intoxicated patients. *Anesthesiology*, **61**(1): 35.
- Goldfrank, L.R. (1994). Toxicologic Emergencies, 5th ed., Appleton & Lange, Norwalk, CT.
- Holme, A.A., Dahlin, D.C., Nelson, S.D. and Dybing, F. (1984). Cytotoxic effects of N-acetyl-p-benzoquinone imine, a common arylating intermediate of paracetamol and N-hydroxyparacetamol. *Biochem. Pharmacol.* **33**: 401406.
- Jaraczewska, W. and Kotwica, M. (1997). Acute poisoning with drugs. A review of the data collected at the national poison information center during the period 1991-1995. *Przegl Lek.* **54**(10): 737-740.
- Khan, M. and Reza. H. (1998). Benzodiazepine self-poisoning in Pakistan: implications for prevention and harm reduction. *J. Pak. Med. Assoc.*, **48**(10): 293-2.
- Lim, A.G. (1989). Death after flumazenil [letter]. *BMJ*, **299**: 858-859.
- Mora, C.T., Torjman, M. and White, P.F. (1989). Effects of diazepam and flumazenil on sedation and hypoxic ventilatory response. *Anesth Analg*, **68**: 473-478.
- Prescott, L.F. and Matthew, H. (1974). Cysteamine for paracetamol over dosage. *Lancet.* **1**: 18.
- Shalansky, S.J., Nauman, T.L. and Englander, F.A. (1993). Therapy update: Effect of flumazenil on benzodiazepine-induced respiratory depression. *Clin. Pharm.* **12**: 483-487.
- Tirmenstein, M.A. and Nelson, S.D. (1989). Subcellular binding and effects on calcium homeostasis produced by acetaminophen and a non-hepatotoxic regioisomer, 3-hydroxyacetanilide in mouse liver. *J. Biol. Chem.* **264**: 9814-9819.
- Trazepacz, P.T., Leavitt, M. and Ciongoli, K. (1992). An animal model for delirium. *Psychopharmacology*. **33**(4): 404-415.